

ICE 3107: Digital Signal Processing

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Lecture Outcomes

- **Develop** familiarity with Linear Time-Invariant (LTI) systems.
- **Compute** convolution and correlation to analyze input–output relationships in digital signal processing (DSP) systems.

Linear Time-Invariant (LTI) System

A Linear Time-Invariant (LTI) system is a fundamental concept in engineering and mathematics used to model systems whose output is predictable based on two specific properties: linearity and time invariance.

An LTI system must satisfy two defining criteria:

- Linearity: The system obeys the principles of superposition.
- Time Invariance: The system's characteristics do not change over time.

LTI System (Cont.)

- For an LTI system, the impulse response completely describes the system behavior.
- Impulse response is the output of a system when the input is an impulse signal. Impulse response tells: "How does a system react to a single instant shock?"

- We denote it by:

$$h(n)$$

- So:

$$h(n) = \textit{output when input } x(n) = \delta(n)$$

- Impulse signal (unit impulse) is:

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

LTI System (Cont.)

- For any LTI system:

$$\text{Output} = \textit{Input} * \textit{Impulse Response}$$

$$y(n) = x(n) * h(n)$$

Output is convolution.

It means: The output of an LTI system is built by combining many small responses of the system.

LTI System (Cont.)

- **Step 1: Break input into impulses**
  - Any signal can be written as: Many shifted impulses
  - So: input = sum of small spikes

- **Step 2: System responds to each impulse**
  - impulse $\rightarrow h(n)$
  - shifted impulse $\rightarrow$ shifted $h(n-k)$
  - So: each small spike produces a copy of $h(n)$

- **Step 3: Add all responses**
  - All shifted responses are added together.
  - That final sum is the output.

Convolution

Final Convolution Formula:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

Where,

- n = output time index
- k = running (dummy) index for input samples

Convolution: It is a mathematical operation that combines an input signal with a system's impulse response to produce the output of a Linear Time-Invariant (LTI) system.

Convolution (Cont.)

Steps in Convolution

- **Folding**

➢Fold $h(k)$about $k = 0$ to obtain $h(-k)$.

- **Shifting**

➢Shift $h(-k)$ by n to the right (to the left if n is negative) to obtain $h(n-k)$.

- **Multiplication**

➢Multiply $x(k)$ by $h(n-k)$to obtain the product sequence:

➢$v_n(k) = x(k)\, h(n-k)$

- **Summation**

➢Sum all values of the product sequence $v_n(k)$to obtain the output at time n.

Convolution (Cont.)

Find the convolution:

1. If x(n)=[1,2] and h(n)=[1,2]

Solution:

Output length L=2+2-1=3

y(0), y(1), y(2)

y(0):	**y(1):**	**y(2):**
• k=0: x(0) h(0-0)=1*1=1 • k=1: x(1) h(0-1)=not exist • y(0)=1	• k=0: x(0) h(1-0)=1*2=2 • k=1: x(1) h(1-1)= 2*1=2 • y(0)=2+2=4	• k=0: x(0) h(2-0)=not exist • k=1: x(1) h(2-1)=2*2=4 • y(1)=4

Convolution (Cont.)

Find the convolution:

2. If x(n)=[1,2↑,3] and h(n)=[1,2,1]

Solution: y(n)=[1, 4↑, 8, 8, 3]

Correlation

Correlation measures the similarity between two signals as a function of a time shift (lag).

It answers the question: "how much does signal x resemble signal y if I slide one past the other?"

It is one of the most practically important operations in DSP — used in radar, sonar, communications, image processing, and pattern recognition

Correlation

Type 1 — Cross-correlation: Measures similarity between **two different signals** x and y.

Cross-correlation formula:

$$r_{xy}(n) = \sum_{k=-\infty}^{\infty} x(k)\, y(k+n)$$

TYPE 2 — AUTOCORRELATION: Measures similarity of a signal **with itself** at different lags.

Autocorrelation formula:

$$r_{xx}(n) = \sum_{k=-\infty}^{\infty} x(k)\, x(k+n)$$

Correlation

Find the correlation:

1. If x(n)=[1, 2] and y(n)=[1,1] then $r_{xy}(n)$=?

Solution: $r_{xy}(n)$=[3, 1, 0]

2. If x(n)=[3,2↑] then $r_{xx}(n)$=?

Solution: y(n)=[6, 13↑, 6]

Thank You


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